

Assignment 2—Conditionals and Loops

Due: 6 April

Submit your work as one .zip file on Brightspace.

Each .java file must have your name (and the name of your partner if you work with one) in a comment at the top.

This assignment exercises conditionals and loops. It does *not* need arrays.

Problem 1 (20 points)

Write a Java class *Stars.java* with a main program that inputs a positive integer n from the user and prints n rows of stars. The first row has one star, the second row has two stars, the third row three, and so on up to the n th row having n stars.

A sample run might look like this:

```
Enter a positive integer: 4
*
**
***
****
```

Problem 2 (40 points)

Problems 2 and 3 were inspired by the character Yossarian in Joseph Heller's novel *Catch-22*.

Write a Java class *Vowels.java* that reads user input a line at a time until the user signals end of file. For each line, it prints the line with all vowels removed. Vowels are the characters *a e i o u* in upper or lower case. If a word consists entirely of vowels, simply removing it creates a double space, so convert the resulting double space to a single space.

A sample run might look like this:

```
Enter text, type <ctrl>+d to exit:
I am a pilgrim and a stranger
m plgrm nd strngr
Traveling through this wearisome land
Trvlng thrgh ths wrsm lnd
<ctrl>+d
```

You can go through each line a character at a time (**charAt** function) and check whether it is a vowel before appending it to the result.

Problem 3

Write a Java class *Articles.java* that reads user input a line at a time until the user signals end of file. For each line, it prints the line with the articles “a”, “an” and “the” removed. As in Problem 2, don’t print double spaces that result from word removal.

A sample run might look like this:

```
Enter text, type <ctrl>+d to exit:
A day at the races
day at races
The best pet is not a raccoon.
best pet is not raccoon.
<ctrl>+d
```

For this problem you need to search the input line for substrings. We mentioned in lecture that the `String` class has a function **indexOf** that will search a string for a particular character. That function can also be used to search for substrings: if **s1** and **s2** are strings, **s1.indexOf(s2)** will return the index in **s1** of the first occurrence of **s2**, or -1 if **s2** does not appear in **s1**.